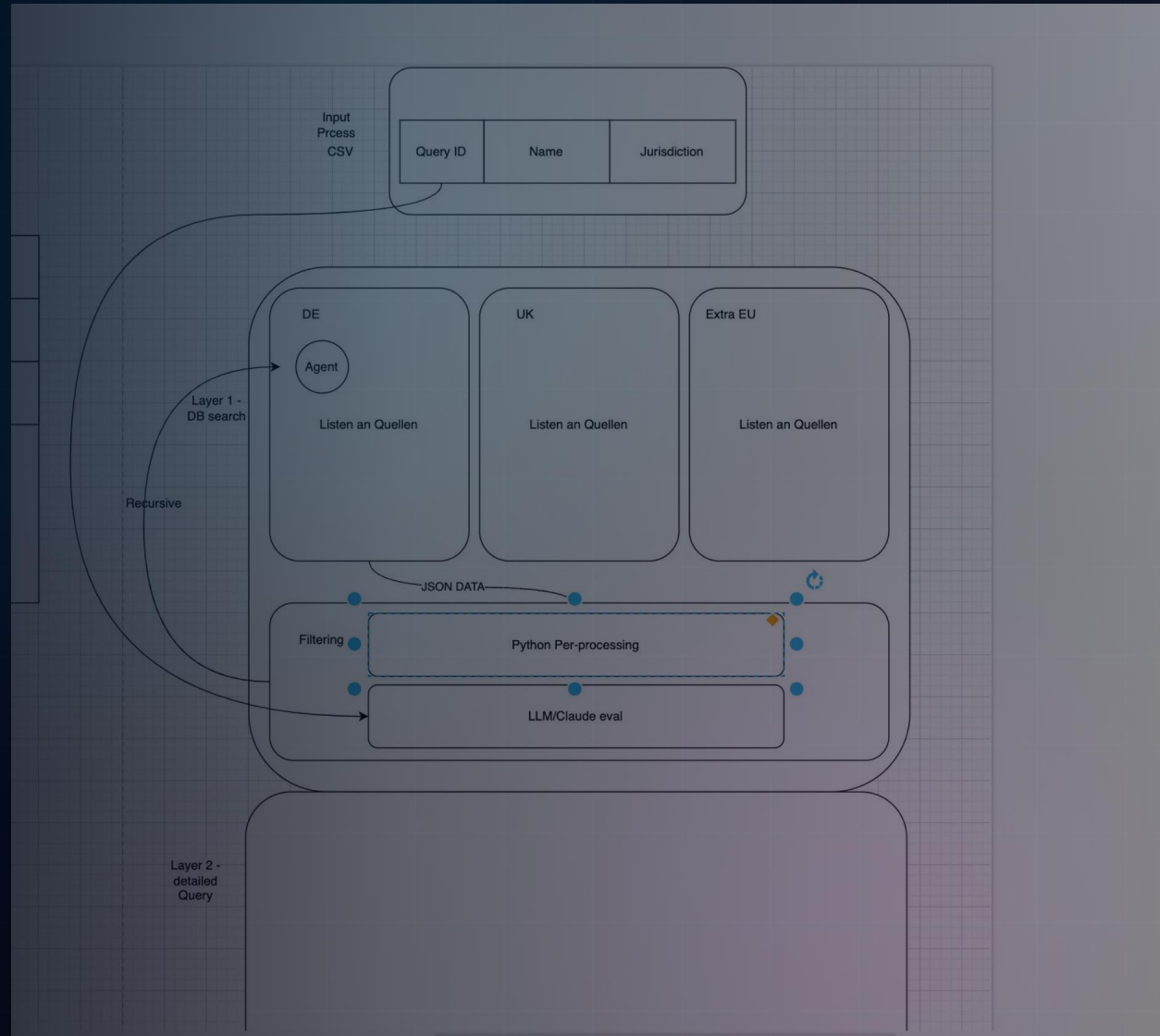


- FEDERATED KYB IDENTIFICATION

Precision over Recall

A federated, LLM-powered KYB pipeline built for precision and full transparency.



THE CORE CHALLENGE

KYB Starts With One Precise Question



Inputs are messy — **partial names**, **transliterations**, wrong jurisdictions



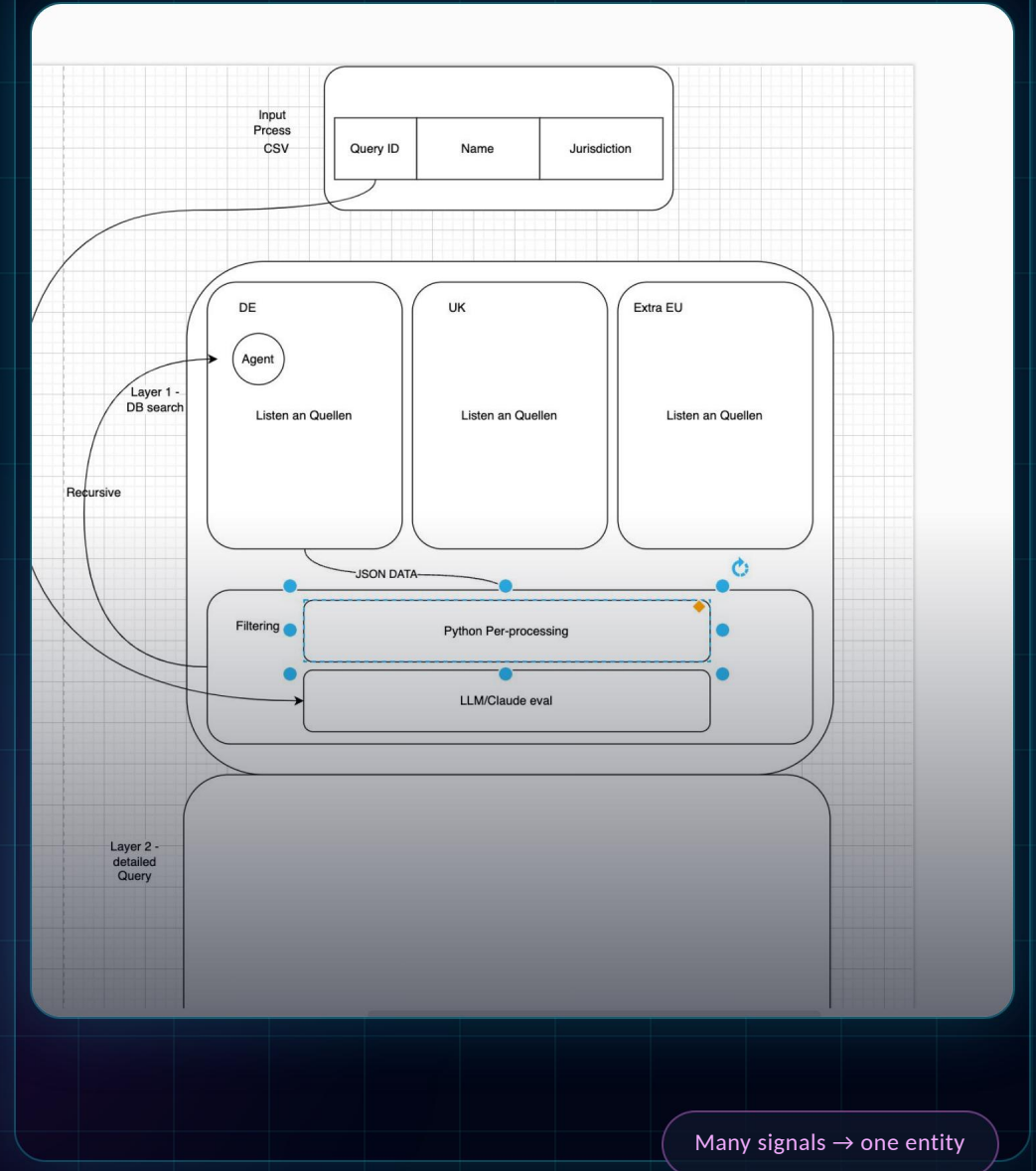
Registries are **fragmented** with no shared schema



Index lookups **lag reality**; direct queries are slow



A confident **wrong ID** is the worst possible outcome



OUR CORE PRINCIPLE

We Choose Honest Over Guessing

We abstain from uncertain identifications — prioritizing precision and full-source traceability over risky guesses.



Precision beats recall



Hallucination is penalized



Blank-but-honest is neutral



Confidence + source wired throughout



ARCHITECTURE

A Two-Layer, Recursive Pipeline

Process CSV inputs

Each query parsed per row: name + jurisdiction.

Parallel registry agents

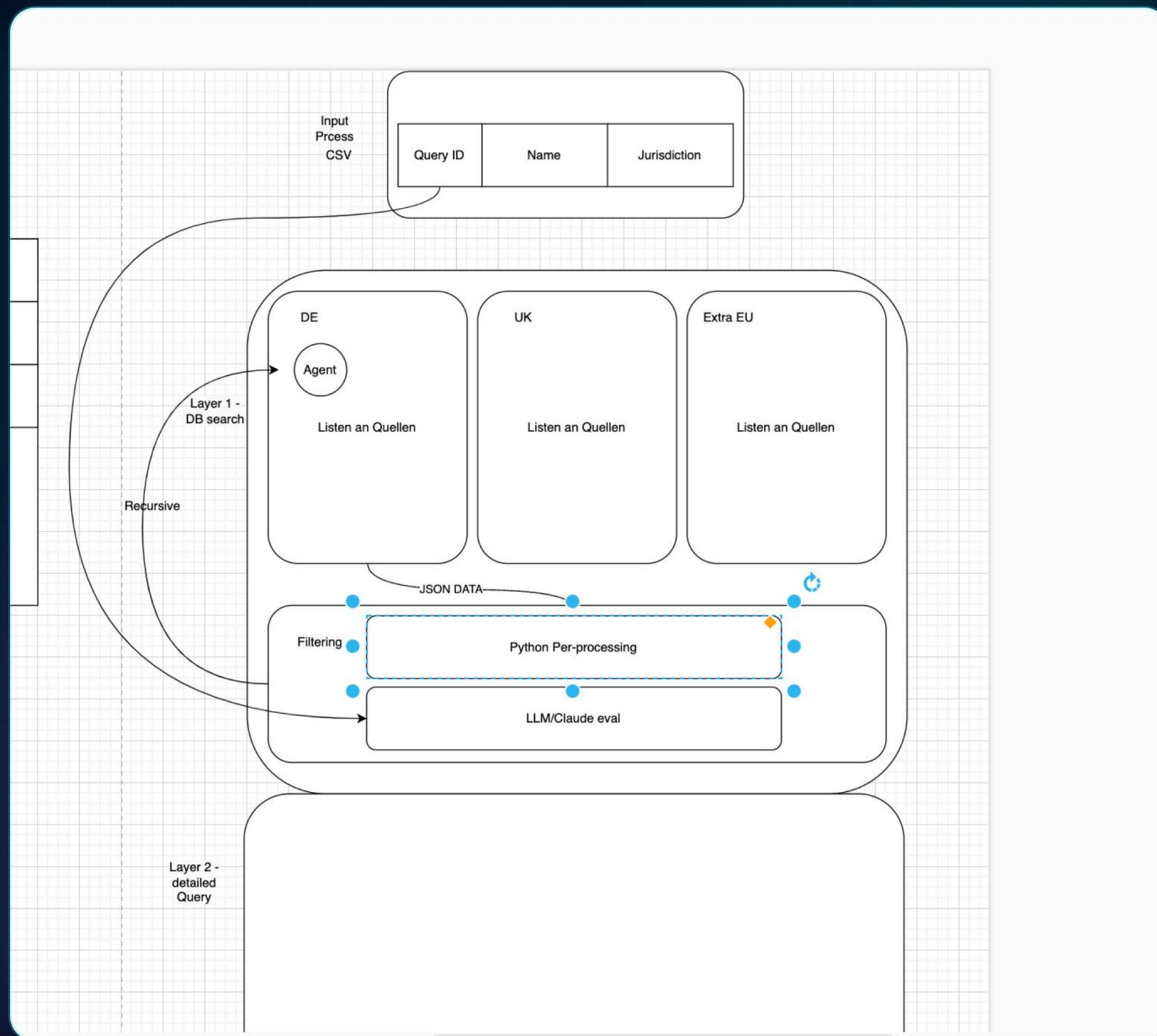
DE, UK & Extra-EU searched simultaneously.

Python + LLM evaluation

Filter, rank & score candidate matches.

Enrichment + feedback loop

Layer 2 deepens detail, recursively refines.



Federated Registry Search

Parallel jurisdiction agents retrieve structured candidates from trusted registries.

3 parallel jurisdiction agents

DE Germany

Handelsregister via curated court/source list.

Registergericht-aware

UK United Kingdom

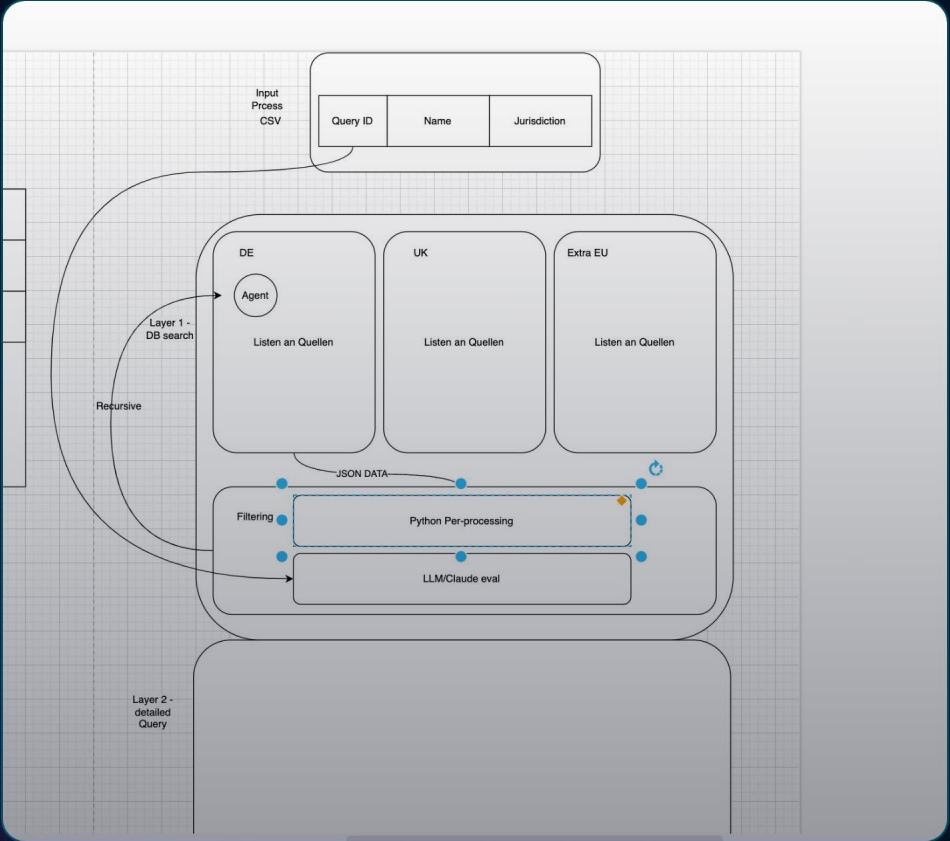
Companies House — free, well-documented API.

End-to-end first

EU+ Extra-EU

GLEIF, OpenCorporates, BRIS as signals.

Extendable



● Curated source lists ● Structured JSON candidates ● New regions plug in easily

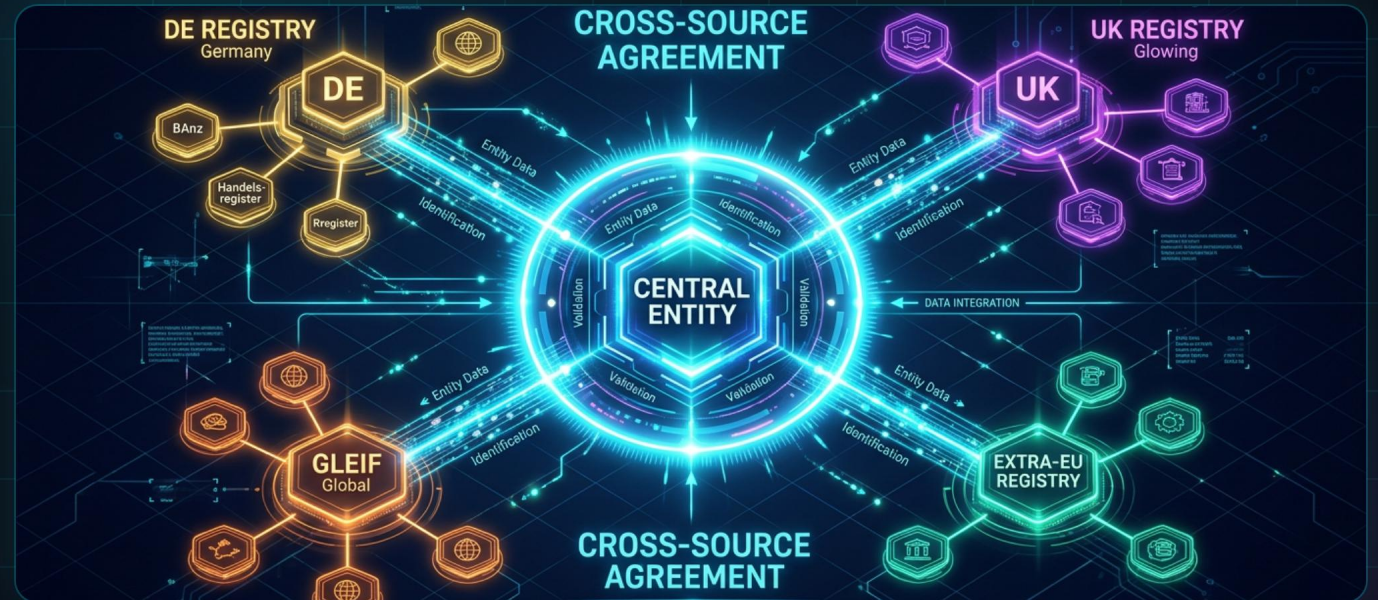
Source: federated agents emit JSON candidates → Python pre-processing → LLM/Claude evaluation.

Fame — Corroboration as a Signal

XN

independent sources agreeing

- Counts registries returning the same entity
- More sources → more likely correct



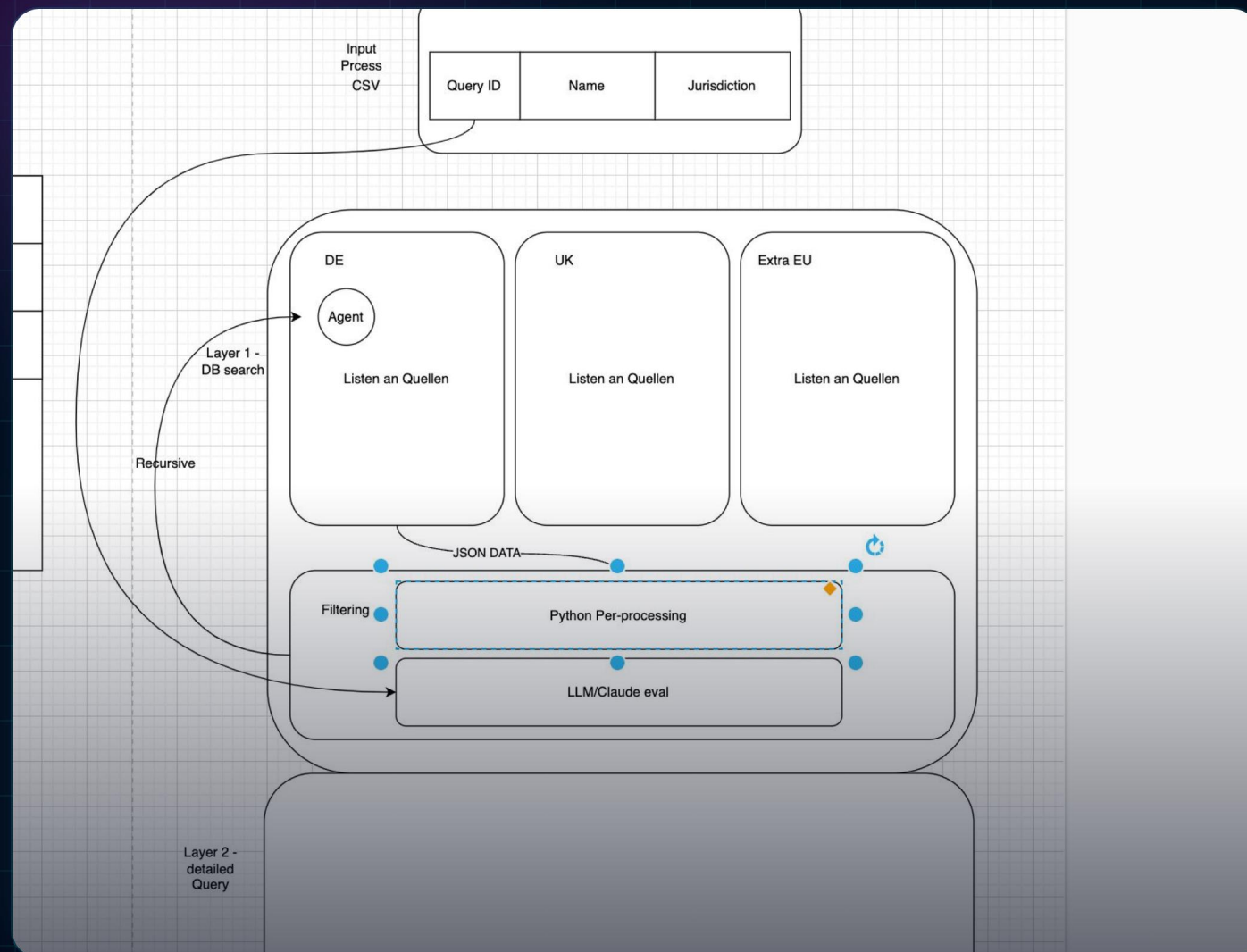
Fame biases the ratio

Corroboration nudges the confidence score upward.

Boosted pre-LLM

Well-corroborated entities rise before reasoning.

From Candidates to a Calibrated Answer



Python pre-processing

Normalizes raw registry results into a clean candidate set.



LLM ranks with calibrated confidence

Claude scores each match in $[0, 1]$.



Every claim cites its source

Citable registry URL backs each field.



Top match — or honest no-match

We abstain rather than invent an ID.

RapidFuzz — A Heuristic Pre-Filter

A deterministic, token-free filter rapidly narrows candidates before LLM reasoning.

4

SCORING PARAMETERS

Fast · Deterministic · No LLM cost

1

Word ratio

Penalizes extra / missing tokens between names

2

Prefix & suffix stripping

Removes legal forms before scoring

3

Fellegi-Sunter TF weighting

Rare tokens weighted higher, via Splink

4

Containment bonus

Rewards subset names within candidates

RapidFuzz — A Heuristic Pre-Filter

4

scoring parameters

deterministic · zero tokens · explainable

A fast, fully explainable filter narrows candidates **before** any LLM runs.

01 Word ratio

Penalizes extra and missing tokens

02 Prefix & suffix stripping

Normalize legal forms before scoring

03 Fellegi-Sunter TF weighting

Rare tokens weighted higher (Splink)

04 Containment bonus

Rewards subset / substring names

Detailed Query & Enrichment

Enrichment enhances verified entity matches with deeper, tiered data.

A

Tier A — Core Attributes

Registered address · status · legal type · VAT number

B

Tier B — People

Officers · directors · legal representatives · signatories

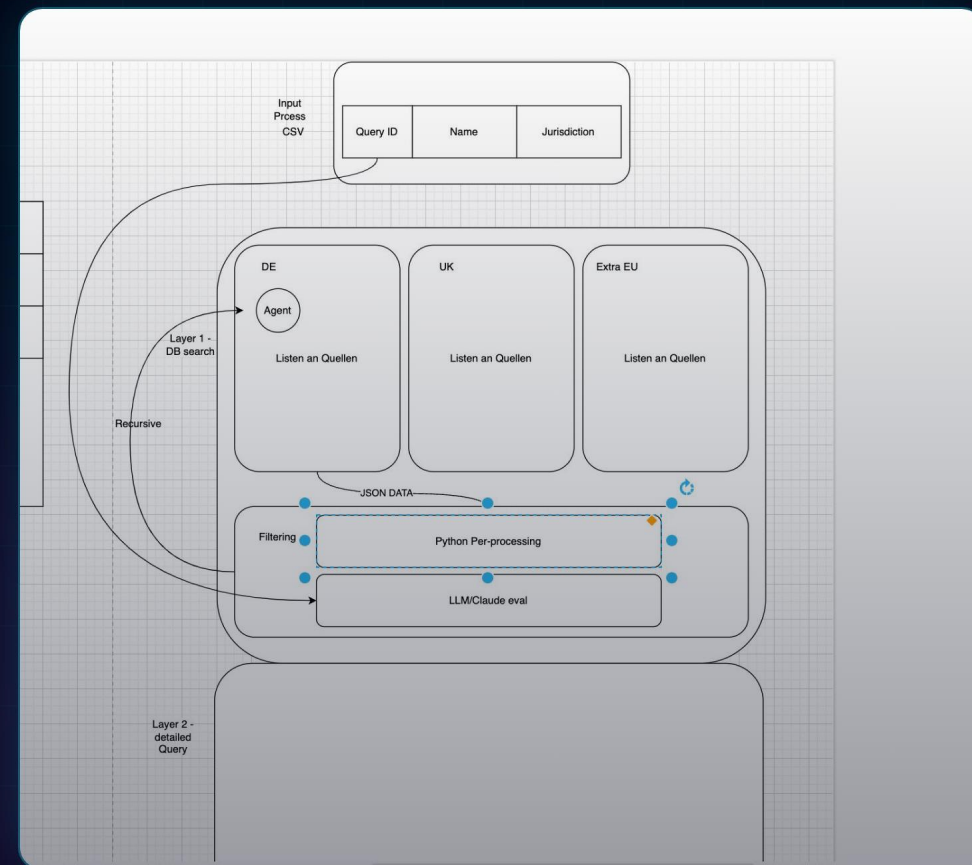
C

Tier C — UBOs

Beneficial owners — honest by design, *never guessed*



Recursive loop refines weak matches before enriching



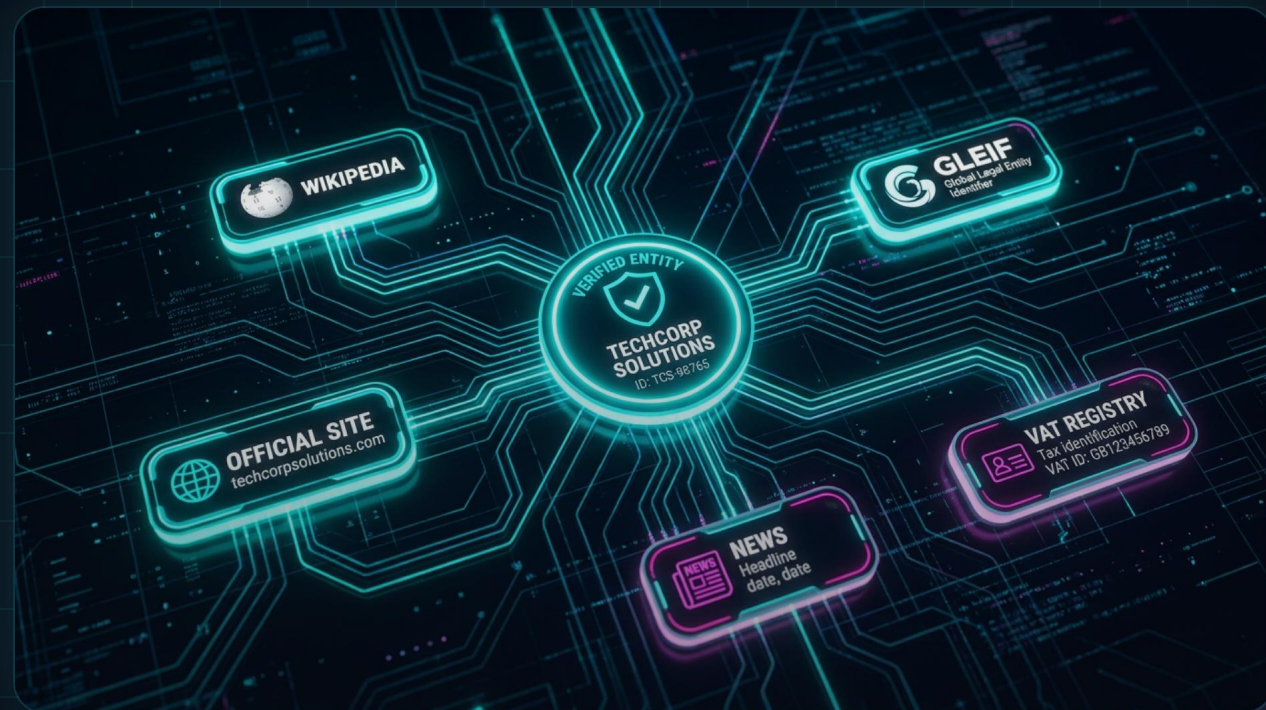
• OPTIONAL LAYER • LAYER 2+

Enrichment Agent

Filling the Gaps

- 1 Fires on 100% ties or hard fields
- 2 Wikipedia, official sites, registers, LEI/VAT
- 3 Fetches CEO, capital, missing attributes
- 4 Every source returned **with a URL**

Optional layer • adds latency • all claims source-linked.



Hallucination Economics Drive Honest Outputs

2/4

of the rubric rewards **abstention & reasoning** — honesty scores as high as a correct match.



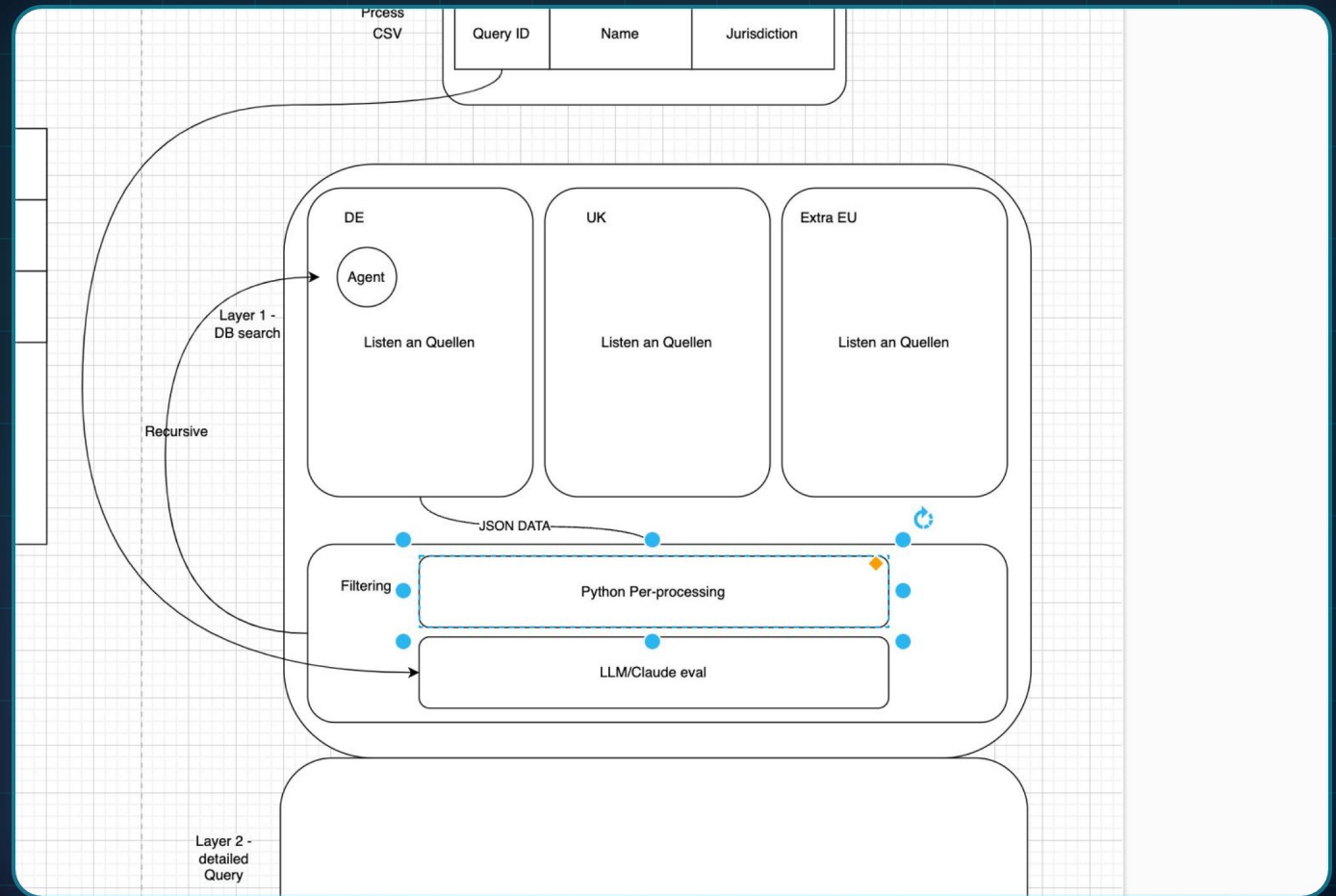
Every claim cites a **verifiable source**



Calibrated confidence, not false certainty



no_match_reason instead of invented IDs

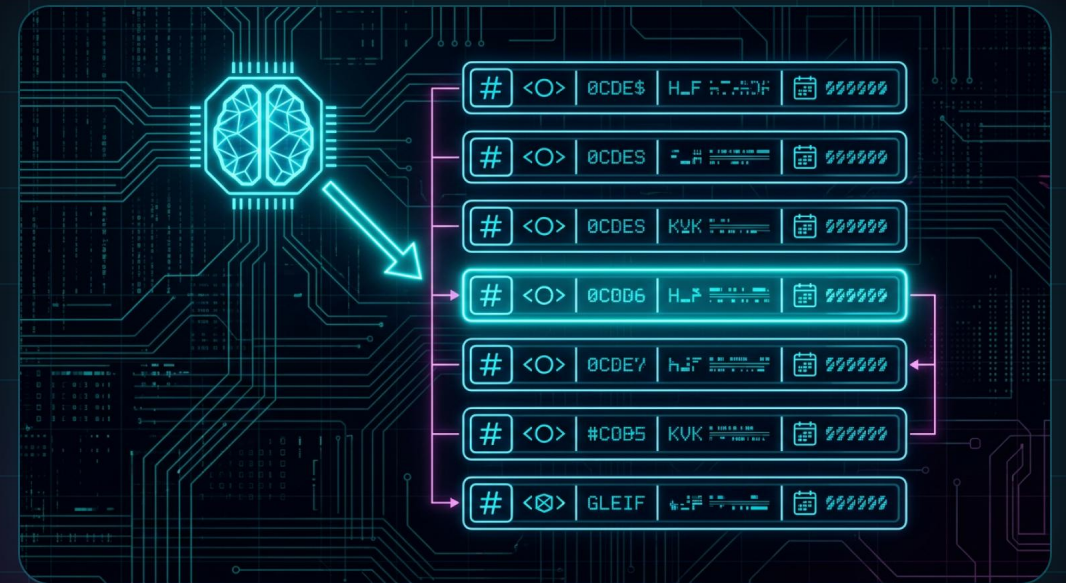


ANTI-HALLUCINATION GUARANTEE

How We Guarantee No Hallucination

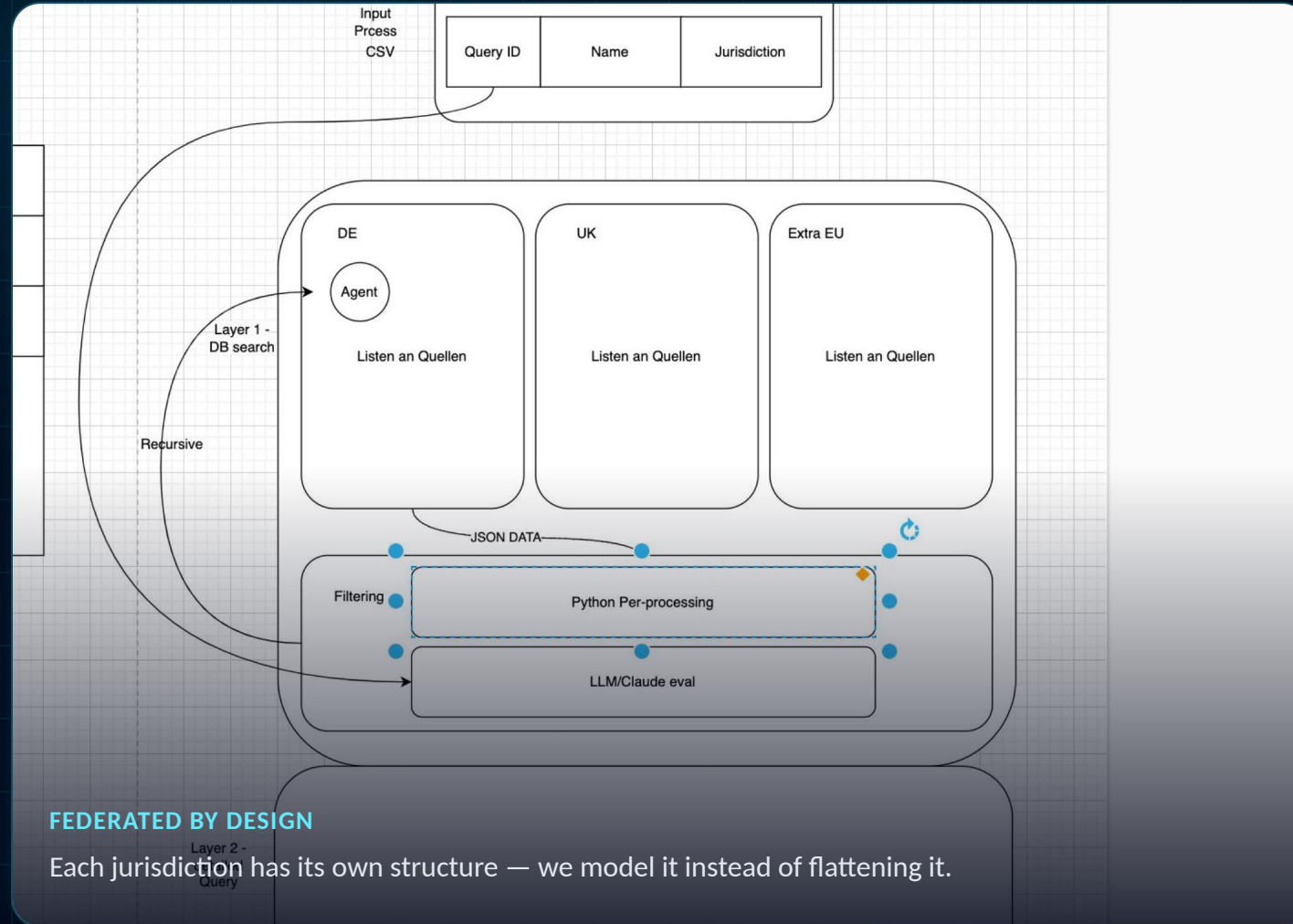
Each entity ID is **selected** from official registry data — never invented by the model.

- 1 Model returns an **index** into real records
- 2 Candidates only from official registry APIs
- 3 Confidence = **fuzz + fame + LLM**, not LLM alone
- 4 Below threshold → honest IDK + recursive retry



LLM selects **record[i]** — it cannot author new IDs.

Respecting how registries **actually** work



DE

Always return Registergericht

DE/AT IDs are court-scoped — HRB 12345 differs per Amtsgericht.

—

HRA vs HRB distinction

Partnerships vs corporations resolved correctly, never conflated.

—

Sole proprietors → honest blank

No registry entry? Blank registry_id with no_match_reason — never invented.

🔒

EU UBO access restricted (2022)

Post-ECJ ruling, UBO data handled honestly — not guessed.

AUTOMATION

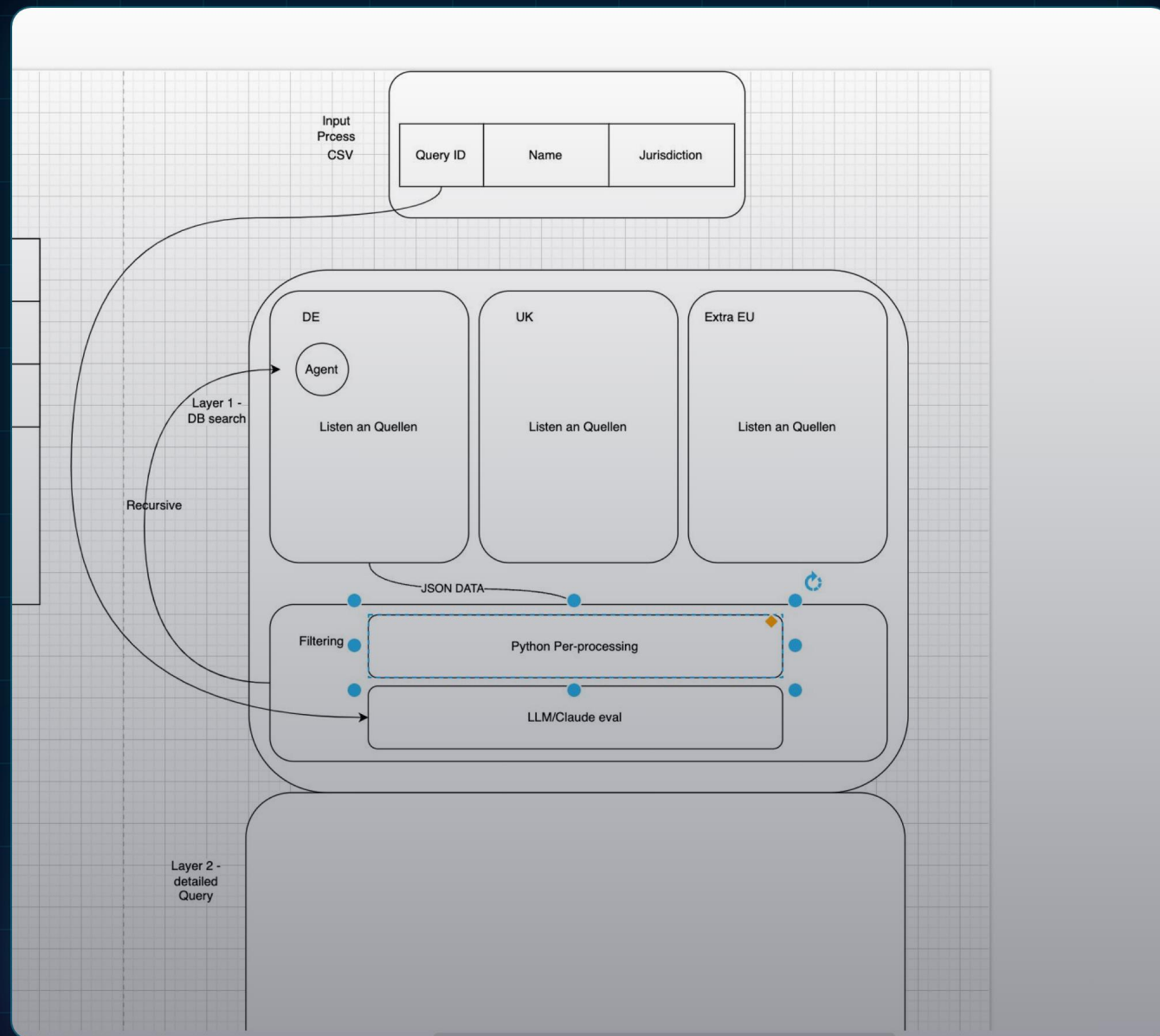
CSV In, Result CSV Out

No manual steps. The pipeline runs end-to-end.

8 output fields
per query

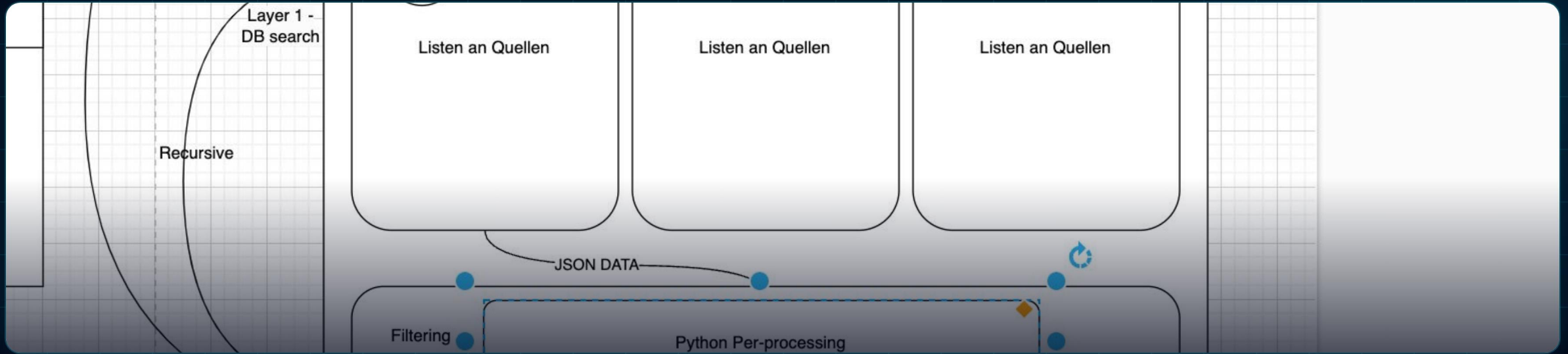
- ✓ Fully automated end-to-end run
- ✓ UI makes the system "think" visibly
- ✓ Live queries, candidates, confidence & sources

Output fields: registry_id, court, normalized name, jurisdiction, confidence, source, no_match_reason.



- A transparent, interactive UI — built to show reasoning, not just move files.

What Our Approach Delivers



Auditable

Every claim citable — compliance can verify, not redo.



Federated

Extensible across jurisdictions and registries.



Calibrated

Confidence scoring prevents costly mistakes.



Honest

Transparent abstention where no data exists.

Verifiable. Calibrated. Honest.

Recursive

Layer 1 -
DB search

DE

Agent

Listen an Quellen

UK

• OUR KYB SOLUTION

Listen an Quellen

Extra EU

Listen an Quellen

The right answer — or a transparent, source-cited no-match. Never a confident guess.

JSON DATA

Right answer or honest abstention

Trust built on source-citation

Transparent confidence, never guesswork

LLM/Claude eval